00-S-256/14A
June 13. 1986
SUPERSEDING
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FEDERAL SPECIFICATION SHEET

SEWING MACHINE, INDUSTRIAL; FLATBED,

SINGLE NEEDLE, 308 STITCH, PARACHUTE REPAIR

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal Agencies.

The complete requirements for procuring the sewing machines described herein shall consist of this document and the issue in effect of Specification 00-S-256.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the specific requirements for single needle, flatbed, 308 stitch, industrial sewing machines used in the fabrication and repair of aerial delivery equipment.

1.2 Classification. The sewing machines shall be of the following classes as specified (see 6.2).

Class 1 - Medium duty
Class 2 - Heavy duty

2. APPLICABLE DOCUMENTS

2.1 Government publications, The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

6-1-9364 Sewing Machine, Industrial, Flatbed, Single Needle, 308 Stitch, Medium Duty: Parachute Repair and Fabrication Test Requirements

AMSC N/A FSC 3530

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6-1-9365 Sewing Machine, Industrial, Flatbed, Single Needle, 308 Stitch, Heavy Duty: Parachute Repair and Fabrication Test Requirements

(Copies of Military specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Class 1, machine. The class 1 machine shall be reversible with the capability of sewing light cotton and nylon duck, light to medium tape and webbing, cords ranging from 3/32 to 3/16 of an inch in diameter and a combination of any of the above, utilizing metric needle sizes 110, 120, 125, 130, 140, and 180. The machine shall operate at speeds not less than 2000 or greater than 3000 stitches per minute. The machine shall be manually lubricated. The mounting shall be fully submerged. The machine shall be equipped with a bobbin winder and a knee lifter.

3.1.1 Class 1, machine feed. The class 1 machine shall have a reversible type A feed. The presser foot shall be hinged. The feed dog shall have six rows of teeth. They shall include one full row of teeth on each side of the needle opening and two partial rows in the front and two partial rows in the rear of the needle opening. The teeth shall be spaced a minimum of 16 teeth to the inch.

3.1.2 Class 1, presser feet. In addition to the standard presser foot required in 3.1.1, the class 1 machine shall be furnished with grooved presser feet as specified on Drawing 6-1-9364.

3.1.3 Class 1, throat plate. The class 1 machine shall be supplied with a heavy duty throat plate. The throat plate shall be alloy steel and air hardened to a Rockwell C 48-56.

3.1.4 Class 1, stitches. The class 1 machine shall produce a minimum of 5 to an unlimited number of stitches per inch.

3.1.5 Class 1, width of the stitch. The class 1 machine shall produce a maximum stitch width of 3/8 of an inch,

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3.1.6 Class 1, height under the presser foot. The class 1 machine shall provide a minimum height of 15/64 of an inch under the presser foot.

3.1.7 Class 1, working room to the right of the needle. The class 1 machine shall be furnished with a working room distance to the right of the needle of 10 inches minimum to 16 inches maximum.

3.1.8 Class 1, table top. The class 1 machine shall be furnished with a type II table top. The outside dimensions of the table top shall be 20 by 40 by 2 inches thick with a ± 1/4 of an inch tolerance for each dimension.

3.1.9 Class 1, spare parts. The class 1 machine shall be furnished with the following spare parts in the quantity indicated:

Spare parts for class 1 machines

Item identification	Quantity
Face plate screw	2
Tension post	1
Thread take up spring	1
Thread tension spring	1
Thread tension thumb nut	1
Sewing hook assembly	1
Sewing hook holding screw	1
Bobbin, metal	10
Bobbin, case	1
Presser foot	1
Presser foot holding screw	1
Bed slide plate spring	1
Bed slide plate spring screw	1
Tension release pin	1
Feed dog	2
Needle throat plate	2
Needle throat plate screw	2
Needle bar pinch screw	1
Needle bar thread guide	1
Needle set screw	2
Needle, round point, chrome finish	
Metric sizes:	
110	100
120	100
125	100
130	100
140	100
180	100

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3.2 Class 2, machine. The class 2 machine shall be reversible with the capability of sewing light to medium cotton and nylon duck, light to medium tape and webbing, and 3/32 to 3/16 inches in diameter cords utilizing metric needle sizes 120, 125, 130, 140, and 180. The machine shall operate at speeds not less than 1500 or greater than 2500 stitches per minute. The machine shall be manually lubricated. The mounting shall be fully submerged. Each machine shall be supplied with a bobbin winder and a knee lifter.

3,2.1 Class 2, machine feed. The class 2 machine feed shall be equipped with a reversible type A feed and a walking presser foot. The walking presser foot shall consist of an inside foot and an outside foot. The feed dog shall have four rows of teeth. They shall include one full row on each side of the needle opening and two partial rows in front of the needle opening. The teeth shall be spaced a minimum of 16 teeth to the inch.

3.2.2 Class 2, presser feet. In addition to the walking presser foot required in 3.2.1, the class 2 machine shall be furnished with grooved presser feet as specified on Drawing 6-1-9365.

3.2.3 Class 2, throat plate. The class 2 machine shall be supplied with a heavy duty throat plate. The throat plate shall be alloy steel and air hardened to a Rockwell C 48-56.

3.2.4 Class 2, stitches. The class 2 machine shall provide a minimum of 5 to an unlimited number of stitches per inch.

3.2.5 Class 2, width of the stitch. The class 2 machine shall produce a maximum stitch width of 3/8 of an inch.

3.2.6 Class 2, height under the presser foot. The class 2 machine shall provide a minimum height of 25/64 of an inch under the presser foot.

3.2.7 Class 2, working room to the right of the needle. The class 2 machine shall be furnished with a working room distance to the right of the needle of 10 inches minimum to 20 inches maximum.

3,2.8 Class 2, table top. The class 2 machine shall be furnished with a type 11 table top. The outside dimensions of the table top shall be 20 by 40 by 2 inches thick with a ± 1/4 of an inch tolerance for each dimension.

3.2.9 Class 2, spare parts. The class 2 machine shall be furnished with the following spare parts in the quantity indicated:

Spare parts for class 2 machine

Item Identification	Quantity
Face plate screw	2
Tension release pin	1
Thread take up spring	1
Thread tension stud	1

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Spare parts for class 2 machine (cent'd)

Item identification	Quantity
Thread tension thumb nut	1
Sewing hook assembly	1
Sewing hook holding screw	1
Bobbin, metal	10
Bobbin, case	1
Presser foot, inside (ungrooved)	1
Presser foot holding screw, inside	1
Bed slide plate spring	1
Bed slide plate holding screw	1
Feed dog	2
Feed dog screw	2
Needle throat plate	2
Needle throat plate screw	2
Needle bar thread guide, lower	1
Needle set screw	1
Presser foot, outside (ungrooved)	1
Presser foot, outside, holding screw	1
Needle, round point, chrome finish	
Metric sizes:	
120	100
125	100
130	100
140	100
180	100

3.3 Performance.

3.3.1 Class 1 machine, operation. The class 1 machine shall be capable of fabricating, replacing and installing lines, inversion nets and vee tabs on personnel and cargo parachutes when tested as specified in 4.1.1 and 4.2.1. The machine shall show no signs of stalling, pounding, shearing of threads, failure to lock stitches, or deformation of any part during fabrication of these articles.

3.3.2 Class 2 machine, operation. The class 2 machine shall be capable of fabricating, repairing and installing lines and retaining loops on cargo parachutes when tested as specified in 4.1.2 and 4.2.2. The machine shall show no signs of stalling, pounding, shearing of threads, failures to lock stitches, or deformation of any part during the fabrication of these articles.

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4. QUALITY ASSURANCE PROVISIONS

4.1 First article testing. When a first article is required it shall be tested in accordance with the applicable test method specified in table I.

TABLE I. First article tests

Sewing machine, class	Requirement paragraph	Test method
1	3.3.1	4.1.1
2	3.3.2	4.1.2

4.1.1 Class 1 machine test. The class 1 machine shall be operated to fabricate and assemble all of the typical parachute details as shown on Drawing 6-1-9364. Any nonconformance with the requirements of 3.3.1 shall constitute failure of this test.

4.1.2 Class 2 machine test. The class 2 machine shall be operated to fabricate and assemble all of the typical parachute details as shown on Drawing 6-1-9365. Any nonconformance with the requirements of 3.3.2 shall constitute failure of this test.

4,1.3 Certificate of compliance. Certification that the first article submitted by the manufacturer of the heavy duty throat plate meets the hardness requirements of 3.1.3 or 3.2.3 shall be furnished and be acceptable to the Government representative.

4.2 End item testing. Each end item shall be tested in accordance with the applicable test method specified in table II. Failure of any test shall be cause for rejection of the end item involved.

TABLE II. End item testing

Sewing machine, class	Requirement paragraph	Test method
1	3.3.1	4.2.1
2	3.3.2	4.2.2

4.2.1 Class 1 machine test. Each class 1 machine shall be operated to sew samples of typical parachute construction, as shown by details F and G on Drawing 6-1-9364. Any nonconformance with the requirements of 3.3.1 shall constitute failure of this test. The completed test samples shall be supplied with each machine furnished.

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4.2.2 Class 2 machine test. Each class 2 machine shall be operated to sew samples of typical parachute construction, as shown by details F and G on Drawing 6-1-9365. Any nonconformance with the requirements of 3.3.2 shall constitute failure of this test, The completed test samples shall be supplied with each machine furnished.

4.2.3 Certificate of compliance. Certification that the production end items by the manufacturer of the heavy duty throat plate meets the hardness requirements of 3.1.3 or 3.2.3 shall be furnished and be acceptable to the Government representative.

5. PACKAGING

Packaging shall be as specified in the basic specification and in the contract or purchase order.

6. NOTES

6.1 Intended use. The sewing machines are intended for industrial use for fabricating and repairing parachutes.

6.2 Ordering data.

a. Title, number and date of this specification.
b. Class of machine required (see 1.2).

MILITARY INTERESTS:

Custodians:

Army - GL
Air Force - 99

Review activity:

DLA - GS

User activity:

Air Force - 50

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

Preparing activity:

Army - GL

Project No. 3530-0079

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